

P3/P4/P5 leads the retained localization and recall columns within this placement-controlled study. This placement result supports the selected multi-scale insertion pattern, while the broader attention-family table above remains non-identical in protocol.

Appendix F - Fourier Configuration and Spectrum Ledger

This appendix separates input-space W1 preprocessing from conventional augmentation and feature-domain Fourier refinement. It preserves the retained candidate screen, transform-order study, timing scope, and the direct paired spectrum analysis.

Table F.1. Retained Fourier candidate screen under the strict control

Configuration	Labeled mAP50	mAP50-95	Healthy FP (%)	Disease miss (%)	HScore (%)
No Fourier control	23.54	6.06	39.02	22.73	14.08
High-pass $\sigma = 50, \alpha = 0.05$	30.88	7.92	34.15	25.00	20.90
W1 high-pass $\sigma = 50, \alpha = 0.10$	34.51	8.85	63.41	19.32	21.08
High-frequency damping	32.11	10.68	39.02	52.27	17.40
Band-pass 12/60, $\alpha = 0.20$	29.52	8.39	26.83	43.18	17.21
Low-frequency flattening	27.87	8.52	14.63	46.59	16.66
Homomorphic filtering	26.12	9.16	41.46	23.86	15.06
Gaussian low-pass $\sigma = 25$	25.08	8.56	31.71	37.50	15.37

W1 uses a Gaussian low-pass mask with $\sigma = 50$ and residual enhancement $\alpha = 0.10$. It changes image intensities without geometric warping, so the associated polygon coordinates remain unchanged. Its effect is augmentation-regime dependent rather than uniformly beneficial.

Table F.2. Offline augmentation-Fourier order ledger

N=1 augmentation	Aug.-only mAP50	Higher Fourier order	Fourier mAP50	Δ (pp)
Translate 0.08	37.82	Augmentation $\rightarrow$ W1	35.00	-2.82
Scale 0.10	36.91	Augmentation $\rightarrow$ W1	31.85	-5.05
FlipLR 0.25	32.75	W1 $\rightarrow$ augmentation	36.51	+3.77
Rotation 3 degrees	31.08	W1 $\rightarrow$ augmentation	35.95	+4.87
Erasing 0.15	19.81	W1 $\rightarrow$ augmentation	29.07	+9.26
FlipUD 0.30	33.45	Augmentation $\rightarrow$ W1	34.51	+1.05

Table F.2 is an offline one-transform order probe. The separately recorded segmentation configuration field erasing=0.15 was inactive in the retained Ultralytics segmentation transform path; consequently, the offline erasing row must not be interpreted as evidence that random erasing executed inside the retained strong online augmentation policy.

Table F.3. Recorded Fourier timing scope

Method	Fourier preprocessing (ms/image)	YOLO prediction (ms/image)	Effective total (ms/image)
W1 high-pass	48.62	38.77	87.38
Erasing 0.10	0.00	12.69	12.69
FlipUD 0.30	0.00	12.69	12.69
Scale 0.10 + Erasing 0.10	0.00	12.82	12.82

Timing used the 129-image grouped test set, CUDA execution, 640-pixel inputs, ten warm-up images, and 119 timed images. These values are not Android runtime and are not interchangeable with classification screening timing.

Paired W1-augmentation spectrum protocol

Table F.4. Direct paired spectrum-study design

Protocol item	Retained setting	Interpretation boundary
Held-out images	24 images: six each from Healthy, BG, WSSV, and WSSV_BG	Balanced four-stratum diagnostic sample.
Matched augmentation draws	12 stochastic realizations per image	Same random realization is paired across processing paths.
Global paired observations	288 per global image metric	Supports within-image spectrum comparison, not an additional model benchmark.
Global metrics	Low/mid/high frequency energy share; radial spectral power; Laplacian variance; mean gradient magnitude	Measures redistribution of spatial/frequency content.
Lesion-region metrics	Lesion-region gradient where a disease mask exists	Healthy images have no lesion mask; lesion-region observation count is therefore lower.
Claim boundary	Seed-42 diagnostic mechanism analysis	Not causal proof and not multi-seed confirmation.

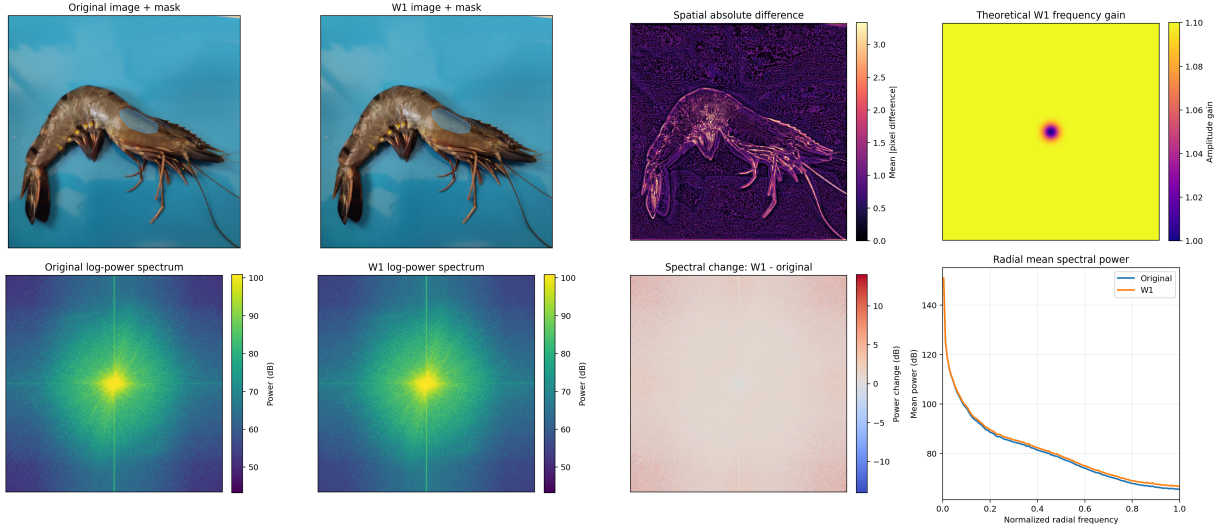


Figure F.1. High-resolution audit view of W1 spatial and frequency-domain behavior at $\sigma = 50$ and $\alpha = 0.10$. The panels show the original and W1 images, spatial difference, theoretical gain, log-power spectra, signed spectral change, and radial mean power.

Figure F.1 documents the preprocessing operator itself: W1 introduces a small residual high-frequency emphasis while preserving image geometry. The theoretical gain panel describes the linear filter before clipping, whereas the image and spectrum panels show the realized effect on a retained shrimp image.

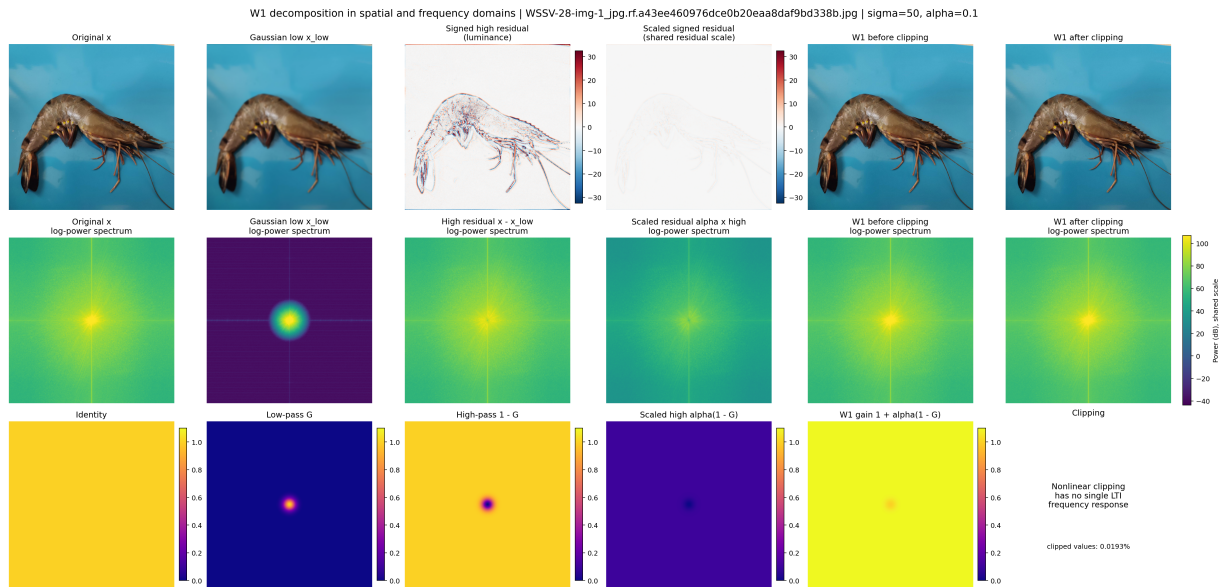


Figure F.2. Supplementary W1 decomposition showing the Gaussian low-pass component, signed high-frequency residual, scaled residual, pre-clipping output, post-clipping output, and corresponding frequency responses.

Figure F.2 separates the linear residual construction from nonlinear clipping. This distinction is important because clipping has no single linear time-invariant frequency response; the reported spectrum analysis therefore evaluates the actual processed images rather than assuming the theoretical gain fully determines the final spectrum.

Table F.5. Retained spectrum-effect summary under strong augmentation

Metric	Retained numerical evidence	Retained fraction	Interpretation
High-band energy share	W1 alone: +0.000670; incremental W1 after strong augmentation: +0.000151	22.5%	Most of W1's global high-band emphasis overlaps with or is redistributed by the strong pipeline.
Laplacian variance	Retained ratio reported from the paired study	21.1%	Edge-energy increment is strongly attenuated after strong augmentation.
Mean gradient magnitude	Retained ratio reported from the paired study	56.1%	A larger fraction of average gradient effect remains than for high-band share.
Lesion-region gradient	Retained ratio reported for disease-mask regions	approx. 79%	Local lesion-region edge emphasis is less attenuated than the global spectral summary.